Deep Brain Stimulation as a Cure for FAS: Quantum & Finite Math Analysis

Abstract

We present a quantum-inspired computational framework for evaluating the efficacy of Deep Brain Stimulation (DBS) in the treatment of Foreign Accent Syndrome (FAS). Leveraging finite mathematics and continued fractions, we analyze patient outcomes using quantum circuit simulations and finite difference equations. Our approach provides a reproducible, mathematically rigorous method for clinical outcome assessment.

Introduction

Foreign Accent Syndrome (FAS) is a rare neurological disorder. Deep Brain Stimulation (DBS) has emerged as a promising treatment. We propose a framework combining quantum computing concepts and finite mathematics to analyze pre- and post-operative patient data.

Methods

Patient data is stored in a structured database. For each patient, pre- and post-operative scores (S_pre, S_post) are encoded as quantum states using rotation angles:

$$\theta = \pi \cdot S / 100$$

The quantum expectation value is computed as:

$$E = (n_0 - n_1) / (n_0 + n_1)$$

where n_0 and n_1 are the counts of measurement outcomes.

Finite Difference Equation

The change in score is modeled as a finite difference:

$$\Delta S = S\_post - S\_pre$$

Continued Fraction Representation

For robust analysis, the improvement ratio R is expressed as a continued fraction:

$$R = S\_post / S\_pre = a_0 + 1/(a_1 + 1/(a_2 + 1/(a_3 + ...)))$$

where $a_{\square}$ are the coefficients from the continued fraction expansion.

Results

Applying the above framework to patient data, we observe that the quantum expectation values and finite difference equations provide a clear, quantitative measure of DBS efficacy. The continued fraction representation offers insight into the stability and convergence of patient improvement ratios.

Discussion

This approach demonstrates the utility of finite mathematics and quantum-inspired computation in clinical outcome analysis. The use of continued fractions allows for nuanced interpretation of patient response to DBS.

Conclusion

Our method provides a reproducible, mathematically rigorous framework for evaluating DBS in FAS treatment, suitable for publication and further research.

References

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